

Cost Function

Knowing $\hat{y} = \sigma(w^T x + b)$ where $\sigma(z) = \frac{1}{1 + e^{-z}}$

Given $\{(x^1, y^1), \dots, (x^m, y^m)\}$, we want $\hat{y}^{(i)} \approx y^{(i)}$

• Loss function

→ used to find how well our $\hat{y}$ is doing.

These two should be as close as possible

One example of a loss function is:

$$L(\hat{y}, y) = \frac{1}{2} (\hat{y} - y)^2 \rightarrow \text{known as square error}$$

But, we don't usually use it because you end up with an optimization problem, it makes gradient descent not possible

In logistic regression, we use:

$$L(\hat{y}, y) = -(y \log \hat{y} + (1-y) \log(1-\hat{y}))$$

$$\begin{aligned} \text{If } y=1: L(\hat{y}, y) &= -\log \hat{y} & \hat{y} \text{ should be large} \\ \text{If } y=0: L(\hat{y}, y) &= -\log(1-\hat{y}) & 1-\hat{y} \rightarrow \text{large} \\ & & \hat{y} \rightarrow \text{small} \end{aligned}$$

Loss function measures only one sample in training set
Cost function measures the performance on the entire training data.

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^i; y^i)$$

Simply, we take the average of all loss functions.